

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5283B

5 Credits: 4

Program Elective

Source-grounded

IoT Security

□ Understand the basics of IoT, sensors, actuators and IoT boards in real time

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5283B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5283B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics And Information security
Course code	5283B
Course title	IoT Security
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4T:0 P:0) Periods per semester:60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

23 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- Understand the basics of IoT, sensors, actuators and IoT boards in real time environment. Learn arduino programming
- Familiarize analog and digital communication
- Understand the Security requirements in IoT

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding and IoT boards in real time environment Understand the syntax of control statements, 14 Understanding	See official syllabus
CO2	operators , I/O functions for problem solving Understand analog and digital communications	See official syllabus
CO3	15 Understanding with arduino	See official syllabus
CO4	Understand the Security requirements in IoT 14 Understanding	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2 2
CO2	CO2 2 2
CO3	CO3 - 1
CO4	CO4 1 3 3 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	real time environment	5	As specified
Module 2	for problem solving	6	As specified
Module 3	Understand analog and digital communications with arduino	6	As specified
Module 4	Understand the Security requirements in IoT	6	As specified

MODULE 1

real time environment

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: real time environment
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking

3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding versus m2m, cps, wot; enabled technologies, networking should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking എന്നിങ്ങനെ definition/meaning എന്നിങ്ങനെ. എന്നിങ്ങനെ എന്നിങ്ങനെ, steps, application എന്നിങ്ങനെ എന്നിങ്ങനെ. Technical terms English-ൽ എന്നിങ്ങനെ എന്നിങ്ങനെ.

▼ 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics,

3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding components, challenges and applications sensors: definition, characteristics, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Understanding components, Challenges and applications Sensors: Definition, Characteristics, definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations

Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, deviations, types-scalar, multimedia, 3 understanding hybrid and virtual; considerations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- ∞ Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations ∞ ∞ definition/meaning ∞ ∞ . ∞ ∞ ∞ , steps, application ∞ ∞ ∞ . Technical terms English- ∞ ∞ ∞ .

▼ Actuators: Definition and Types 3 Understanding Classification of IoT boards:

Actuators: Definition and Types 3 Understanding Classification of IoT boards: is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Actuators: Definition and Types 3 Understanding Classification of IoT boards: using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, actuators: definition and types 3 understanding classification of iot boards: should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Actuators: Definition and Types 3 Understanding Classification of IoT boards: means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ആക്ടുേറ്റേഴ്സ്: ഡിഫിനിഷൻ ആൻഡ് ടൈപ്പ് 3 Understanding Classification of IoT boards: ഏതു ഓർഡറിലും ഡിഫിനിഷൻ/meaning നൽകണം. ഏതു ഓർഡറിലും ഏതു ഓർഡറിലും, steps, application നൽകണം ഏതു ഓർഡറിലും. Technical terms English-ൽ നൽകണം.

▼ **Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions**

Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions is an official topic in Module 1 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual;

Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, microcontroller boards, single board understanding 3 controller, system on chipboard cps, iot versus wot; enabled technologies, networking components, challenges and applications sensors: definition, characteristics, deviations, types-scalar, multimedia, hybrid and virtual; considerations actuators: definition, types-hydraulic, pneumatic, electric, thermal or magnetic, mechanical, soft and shape memory polymers; characteristics classification of iot boards: microcontroller boards, single board controller, system on chipboard understand the syntax of control statements, operators , i/o functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Microcontroller boards, Single board Understanding 3 controller, System on Chipboard CPS, IoT versus WoT; Enabled technologies, Networking components, Challenges and applications Sensors: Definition, Characteristics, Deviations, Types-Scalar, Multimedia, Hybrid and virtual; Considerations Actuators: Definition, Types-Hydraulic, Pneumatic, Electric, Thermal or Magnetic, Mechanical, Soft and shape memory polymers; Characteristics Classification of IoT boards: Microcontroller boards, Single board controller, System on Chipboard Understand the syntax of control statements, operators , I/O functions □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

for problem solving

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: for problem solving
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Introduction to arduino

Introduction to arduino is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction to arduino using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction to arduino should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction to arduino means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ definition/meaning ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ. ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ, steps, application ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ. Technical terms English-ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ ഇൻ്റ്രൊഡക്ഷൻ ടു അർഡൂയോ.

▼ Program elements

Program elements is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Program elements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, program elements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Program elements means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രോഗ്രാം ഘടകങ്ങൾ (Program elements) നിർവ്വചനം/അർത്ഥം (definition/meaning) വിശദീകരിക്കുക. (Explain the definition/meaning). പദങ്ങൾ, ഘട്ടങ്ങൾ, (steps, application) ഉപയോഗം (application) വിശദീകരിക്കുക. (Explain the application). Technical terms English-ൽ (Technical terms in English) ഉൾപ്പെടുത്തുക. (Include them).

▼ statements

statements is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify statements using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, statements should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What statements means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എന്ന പ്രസ്താവനകൾക്ക് നിർവ്വചനം/അർത്ഥം എഴുതുക. നിർവ്വചനം എഴുതുന്നതിൽ, steps, application എന്നിവയെക്കുറിച്ച് വിവരിക്കുക. Technical terms English-ൽ എഴുതുക.

▼ Connecting an external led

Connecting an external led is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Connecting an external led using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, connecting an external led should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Connecting an external led means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇംഗ്ലീഷ് കണക്റ്റിംഗ് അക്സറിയർ ലെഡ്
 ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്
 ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms
 English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

▼ Working with buttons

Working with buttons is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Working with buttons using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, working with buttons should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Working with buttons means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രവർത്തിക്കുന്ന ബട്ടണുകൾ പ്രവർത്തിക്കുന്നതിന്റെ definition/meaning പ്രവർത്തിക്കുന്നതിന്റെ. പ്രവർത്തിക്കുന്നതിന്റെ steps, application പ്രവർത്തിക്കുന്നതിന്റെ. Technical terms English-ൽ പ്രവർത്തിക്കുന്നതിന്റെ.

▼ Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts

Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts is an official topic in Module 2 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects;

Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interrupts 2 understanding contents :programming with arduino: introduction to arduino, features, components, arduino ide, program elements-structure, variables and constants, data types, operators, control statements, loops, functions, arrays, string objects; time, i/o function, display, random numbers, gpio, controlling leds-blinking led without delay, connecting an external led, rgb led, the 7-segment display; working with buttons-connecting a button, button with no resistor, the toggle switch, button to serial, button multiplexing; interrupts should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഇററു Interrupts 2 Understanding Contents :Programming with Arduino: Introduction to arduino, Features, Components, Arduino IDE, Program elements-Structure, Variables and constants, Data types, Operators, Control statements, Loops, Functions, Arrays, String objects; Time, I/O function, Display, Random numbers, GPIO, Controlling LEDs-Blinking led without delay, Connecting an external led, RGB LED, The 7-segment display; Working with buttons-Connecting a button, Button with no resistor, The toggle switch, Button to serial, Button multiplexing; Interrupts ഇററു definition/meaning ഇററു. ഇററു ഇററു, steps, application ഇററു ഇററു. Technical terms English-ഇററു ഇററു.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Understand analog and digital communications with arduino

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand analog and digital communications with arduino
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ Introduction-Serial communication

Introduction-Serial communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction-Serial communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction-serial communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction-Serial communication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ
 ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ definition/meaning ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ. ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ, steps, application ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ. Technical terms English-ൽ ഇൻ്റ്രൊഡക്ഷൻ-സീരിയൽ കമ്മ്യൂണിക്കേഷൻ.

▼ Introduction Parallel communication

Introduction Parallel communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Introduction Parallel communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, introduction parallel communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Introduction Parallel communication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇൻ്റ്രൊഡക്ഷൻ പാരലൽ കമ്മ്യൂണിക്കേഷൻ
 പാരലൽ കമ്മ്യൂണിക്കേഷൻ ന്റെ definition/meaning എഴുതുക. പാരലൽ കമ്മ്യൂണിക്കേഷൻ
 ന്റെ steps, application എഴുതുക. പാരലൽ കമ്മ്യൂണിക്കേഷൻ ന്റെ Technical terms
 English-ൽ എഴുതുക.

▼ Interfacing LCD character display

Interfacing LCD character display is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Interfacing LCD character display using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, interfacing lcd character display should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Interfacing LCD character display means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇം Interfacing LCD character display
 ഇംഗ്ലീഷ് definition/meaning ഇംഗ്ലീഷ്. ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്
 ഇംഗ്ലീഷ്, steps, application ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്. Technical terms
 English-ഇംഗ്ലീഷ് ഇംഗ്ലീഷ്.

▼ UART Communication

UART Communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify UART Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, uart communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What UART Communication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- $\square\square$ UART Communication $\square\square\square\square\square\square\square\square$ $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square$ $\square\square\square\square$ $\square\square\square\square\square\square$, steps, application $\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$ $\square\square\square\square\square\square\square\square$.

▼ I2C Communication

I2C Communication is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify I2C Communication using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, i2c communication should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What I2C Communication means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഇ I2C Communication ഇതിന്റെ നിർവ്വചനം/അർത്ഥം ഇതുകൊണ്ട് എന്തെങ്കിലും ഉദാഹരണം, steps, application ഇതിന്റെ പ്രാധാന്യം ഇതുകൊണ്ട് എന്തെങ്കിലും ഉദാഹരണം. Technical terms English-ഇ ഇതിന്റെ നിർവ്വചനം/അർത്ഥം.

▼ SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing

SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing is an official topic in Module 3 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, spi communication 3 understanding contents : analog and digital communication with arduino: introduction-serial communication, parallel communication, interfacing lcd character display uart communication: uart protocol, serial communication, interfacing esp8266 module i2c communication: i2c protocol, adxl345 module, interfacing bmp180 module spi communication: spi protocol, sd card interfacing, ethernet module interfacing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What SPI Communication 3 Understanding Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
 Contents : Analog and Digital Communication with Arduino: Introduction-Serial communication, Parallel communication, Interfacing LCD character display UART Communication: UART protocol, Serial communication, Interfacing ESP8266 module I2C Communication: I2C protocol, ADXL345 module, Interfacing BMP180 module SPI Communication: SPI protocol, SD card interfacing, Ethernet module interfacing
 definition/meaning . , steps, application . Technical terms English-
 .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Understand the Security requirements in IoT

This module is presented from the official syllabus scope for IoT Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Understand the Security requirements in IoT
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ 2 Understanding Architecture Security Architecture in the Internet of

2 Understanding Architecture Security Architecture in the Internet of is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Architecture Security Architecture in the Internet of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding architecture security architecture in the internet of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding Architecture Security Architecture in the Internet of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding Architecture Security Architecture in the Internet of definition/meaning steps, application Technical terms English-

▼ 3 Understanding Things

3 Understanding Things is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding Things using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding things should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding Things means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Understanding Things
 definition/meaning . steps, application . Technical terms English-
 .

▼ Security Requirements in IoT 2 Understanding Insufficient Authentication /

Security Requirements in IoT 2 Understanding Insufficient Authentication / is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Security Requirements in IoT 2 Understanding Insufficient Authentication / using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, security requirements in iot 2 understanding insufficient authentication / should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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▼ 2 Understanding Authorization Vulnerabilities – Secrecy and Secret-

2 Understanding Authorization Vulnerabilities – Secrecy and Secret- is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Authorization Vulnerabilities – Secrecy and Secret- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding authorization vulnerabilities – secrecy and secret- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Understanding Authorization Vulnerabilities – Secrecy and Secret- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding Authorization Vulnerabilities – Secrecy and Secret- definition/meaning . . . , steps, application Technical terms English-

▼ 2 Understanding Key Capacity

2 Understanding Key Capacity is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Key Capacity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding key capacity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Key Capacity means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Understanding Key Capacity

പ്രധാനപ്പെട്ട പദങ്ങൾ definition/meaning നൽകണം. പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ, steps, application പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ. Technical terms English-ൽ പദങ്ങൾ പദങ്ങൾ പദങ്ങൾ.

▼ **Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations”, CRC Press, 2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 <https://www.arduino.cc/en/IoT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot>**

Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations”, CRC Press, 2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1

<https://www.arduino.cc/en/loT/HomePage> 2

<https://learn.adafruit.com/category/internet-of-things-iot> is an official topic in Module 4 of IoT Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 and Implementations", CRC Press, 2016 Russell, Brian and Drew Van Duren, "Practical Internet of Things Security", Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 <https://www.arduino.cc/en/loT/HomePage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transport encryption 3 understanding security requirements in iot architecture - security in enabling technologies - security concerns in iot applications. security architecture in the internet of things - security requirements in iot - insufficient authentication/authorization - insecure access control - threats to access control, privacy, and availability - attacks specific to iot. vulnerabilities - secrecy and secret-key capacity - authentication/authorization for smart devices - transport encryption - attack & fault trees text / reference t/r book title/author sudeep mishra, anandarupmukherjee and arijit roy, introduction to iot, new t1 delhi: university cambridge press, 2021. (chapter 4) cornel amariei, arduino development cook book, birmingham: packt t2 publishing ltd., 2015. (chapter 2-6) fei hu, "security and privacy in internet of things (iots): models, algorithms, r1 and implementations", crc press, 2016 russell, brian and drew van duren, "practical internet of things security", packt r2 publishing, 2016. online resources sl.no website link 1 <https://www.arduino.cc/en/iot/homepage> 2 <https://learn.adafruit.com/category/internet-of-things-iot> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, "Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1

Aspect	What to prepare
	andImplementations”, CRC Press,2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1 https://www.arduino.cc/en/IoT/HomePage 2 https://learn.adafruit.com/category/internet-of-things-iot means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Transport Encryption 3 Understanding Security Requirements in IoT Architecture - Security in Enabling Technologies - Security Concerns in IoT Applications. Security Architecture in the Internet of Things - Security Requirements in IoT - Insufficient Authentication/Authorization - Insecure Access Control - Threats to Access Control, Privacy, and Availability - Attacks Specific to IoT. Vulnerabilities - Secrecy and Secret-Key Capacity - Authentication/Authorization for Smart Devices - Transport Encryption - Attack & Fault trees Text / Reference T/R Book Title/Author Sudeep Mishra, Anandarupmukherjee and Arijit Roy, Introduction to IoT, New T1 Delhi: University Cambridge Press, 2021. (Chapter 4) Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt T2 Publishing Ltd., 2015. (Chapter 2-6) Fei HU, “Security and Privacy in Internet of Things (IoTs): Models, Algorithms, R1 andImplementations”, CRC Press,2016 Russell, Brian and Drew Van Duren, “Practical Internet of Things Security”, Packt R2 Publishing, 2016. Online Resources Sl.No Website Link 1
<https://www.arduino.cc/en/IoT/HomePage> 2

<https://learn.adafruit.com/category/internet-of-things-iot> എന്നുള്ളതാണ് ഇതിന്റെ definition/meaning എന്നുള്ളതാണ്. ഇതിന്റെ steps, application എന്നിവയെക്കുറിച്ചുള്ള Technical terms English-ൽ എഴുതേണ്ടതാണ്.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking. (3 marks)**

► **Define or explain 3 Understanding components, Challenges and applications Sensors: Definition, Characteristics,. (3 marks)**

► **Define or explain Deviations, Types-Scalar, Multimedia, 3 Understanding Hybrid and virtual; Considerations. (3 marks)**

► **Define or explain Actuators: Definition and Types 3 Understanding Classification of IoT boards:. (3 marks)**

Module 2

► **Define or explain Introduction to arduino. (3 marks)**

► **Define or explain Program elements. (3 marks)**

► **Define or explain statements. (3 marks)**

► **Define or explain Connecting an external led. (3 marks)**

Module 3

► **Define or explain Introduction-Serial communication. (3 marks)**

► **Define or explain Introduction Parallel communication. (3 marks)**

► **Define or explain Interfacing LCD character display. (3 marks)**

► **Define or explain UART Communication. (3 marks)**

Module 4

► **Define or explain 2 Understanding Architecture Security Architecture in the Internet of. (3 marks)**

► **Define or explain 3 Understanding Things. (3 marks)**

► **Define or explain Security Requirements in IoT 2 Understanding Insufficient Authentication /. (3 marks)**

► **Define or explain 2 Understanding Authorization Vulnerabilities - Secrecy and Secret-. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding versus M2M, CPS, WoT; Enabled technologies, Networking.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Introduction to arduino.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Introduction-Serial communication.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **2 Understanding Architecture Security Architecture in the Internet of.**

3-mark explanation: Explain one principle, process or comparison from

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.arduino.cc/en/IoT/HomePage>
- <https://learn.adafruit.com/category/internet-of-things-iot>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5283B. Verify institutional updates against the current official curriculum.